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Transforming medical students' speaking-up behaviors in medical errors: The impact of simulation and personalized debriefing

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ABSTRACT

Introduction: Sharing mental models is essential for high-performance teams, and speaking up is key for exchanging critical insights, especially during medical errors. Understanding how health providers and trainees voice their concerns is crucial for improving speaking-up behavior. This study aims to fill a gap in the literature by examining how medical students speak up when they encounter medical errors and assessing the impact of training on their speaking-up patterns.

Method: A quasi-experimental study involving 146 students, who were divided into two groups, was conducted in Northern Taiwan. One group of students encountered life-threatening scenario before intervention, followed by a faculty-led personalized debriefing session, then a non-life-threatening scenario after the intervention. Another group of students underwent these sessions in the reverse order. Students' Speaking-up patterns, including expression style, form and attitude, and their speaking-up confidence were assessed at pre- and post-intervention scenarios.

Results: During pre-intervention scenario, in expression style, 50 students (34.5%) addressed their concerns to medical errors with direct expression and 14 students (9.7%) utilized indirect hint to express their concerns. In expression form, 31 students (21.4%) addressed their concerns to medical errors with affirmative sentences and 33 students (22.8%) asked questions to express their concerns. In speaking-up attitude, 47 students (32.4%) used unoffensive words, while 17 students (11.7%) used offensive words. After intervention, significantly change of speaking-up styles, forms, and attitude were observed along with their speaking-up confidence ($p < 0.001$).

Discussion: Medical students are inclined to speak up in the event of medical errors using more direct expression and affirmative sentences, along with increased speaking-up confidence after simulation scenario learning and faculty-led personalized debriefing. Healthcare educators can focus more on discussing with students the advantages and disadvantages of various approaches of speaking-up in medical errors, helping them to develop effective speaking-up behaviors in a variety of medical contexts.

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Introduction

Medical errors, spanning various healthcare domains such as diagnosis, drug administration, and surgical procedures, pose significant risks to patient safety [1]. Worldwide healthcare systems are urged to enhance the safety and quality of healthcare services through policy actions [2]. A crucial aspect of mitigating these risks involves healthcare professionals speaking up upon identifying potential patient safety concerns [3]. About 80% of the serious patient safety events were related to miscommunications among healthcare providers [4]. Developing mature team communication skills is pivotal for patient safety [5]. Sharing a mental model is one of the key factors in a high-performance team [6], with speaking up serving as a pivotal method for exchanging critical insights, particularly in the event of medical errors. Understanding speaking-up patterns among health providers or trainees is essential for developing team

Practice points

- Exploring medical students' speaking-up patterns during medical errors enhances learning.
- After simulated medical error scenario and personalized debriefing, medical students increased direct expression and affirmative sentences when addressing medical errors.
- Learning non-offensive language is vital for medical student to adapt effective speaking-up.
- Identifying and assisting medical students who hesitate to speak up is crucial.

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communication skills. Current evidence on this topic is limited, and relevant analyses and training programs are not widely implemented in regular educational curricula.

The choice of language and phrasing that healthcare providers use when addressing medical errors through speaking up is influenced by the individuals involved, the dynamics of the healthcare team, and the specific context of the situation. Structured tools such as the two-challenge rule, concerned, uncomfortable, and safety issue (CUS), and describe, express, suggest, and consequences (DESC) from TeamSTEPPS [7], as well as Probe, Alert, Challenge, and Emergency action (PACE) [8], have been introduced to assist healthcare providers in speaking up to address medical errors. Despite their availability, tailoring these tools to fit providers' personalities and preferred phrases ensures they can apply these skills effectively and consistently in their careers [9]. This warrants that understanding the patterns of speaking-up is crucial for providing effective feedback. In turn, this facilitates the development of healthcare providers' speaking-up behavior and enables the creation of personalized learning activities [10,11].

An often-neglected skillset in educational curricula is challenging authority [12]. Addressing concerns to senior individuals who make mistakes poses a hierarchical challenge for healthcare providers [13,14]. Barriers, including concerns about repercussions, relationship compromises, and insufficient communication skills, hinder healthcare providers from speaking up in the event of medical errors [15,16]. It is crucial to assist healthcare providers in acquiring a range of adequate assertive communication strategies to navigate various clinical scenarios effectively.

Educational strategies to enhance speaking-up capabilities often comprise several pedagogical approaches [17]. Simulation, which presents clinical scenarios to reflect healthcare situations, enables learners to engage in practical experiences, reflecting on and learning from these experiences to apply in future scenarios [18]. Simulation scenarios provide healthcare educators and researchers opportunities to observe and assess the patterns of medical students' behavior of speaking-up when facing medical errors in a controlled and standardized setting, a task that poses significant challenges in the clinical environment. Studies have shown simulation training have positive influences on medical and nursing students' speaking-up attitudes and behaviors in the event of medical error [10,11,19].

Most studies explore the effects of different educational strategies and their impact on learning outcomes related to speaking up about medical errors, as well as the potential inhibitors and enablers of speaking-up [14,20,21]. Research into the specific patterns of speaking-up behavior during medical errors is scarce [21]. This study addresses the literature gap regarding how medical students speak up in the event of medical errors and assessing the impact of training on their speaking-up patterns. It is important for healthcare educators to gain a deeper understanding of these speaking-up patterns in medical error contexts and assess any changes in behavior following training to develop effective, personalized assertive communication programs. This study aimed to explore changes in medical students' speaking-up behavior, specifically in phrase patterns and attitudes, following a simulated scenario and personalized debriefing session. This addresses the gap in understanding the efficacy of assertive communication training. The hypotheses of this study were: (1) speaking-up patterns vary across different clinical conditions; (2) speaking-up patterns differ before and after personalized debriefing training.

Method

Setting and participants

The research team conducted a quasi-experimental study at Taipei Medical University. In this study, we adopted a protocol that mirrors the one from our prior work which was conducted within a speaking-up training course for 153 medical students in 2018 [10,11]. This consistent methodology has proven effective in examining medical students' speaking-up behaviors in the event of medical errors. The study was approved by the institutional review board at Taipei Medical University. This took place at the simulation training center of a medical university in Northern Taiwan, from March to December 2020. A total of 146 medical students, comprising 81 males and 65 females, participated. All medical students were undergoing clerkship rotations in the sixth year of their 6-year medical school curriculum. All students had previously attended mandatory patient safety lectures in the beginning of their orientation course and during their clinical rotation at the hospital. Prior to entering the simulation course, the medical students were certified as advanced cardiac life support providers.

All medical students participated in the speaking up training course during their second clinical internal medicine clerkship rotation. The study spanned 18 weeks during the study period. During each course day, 8–10 participants attended, and administrative staff randomly assigned them to different groups. Both simulation groups operated concurrently within the one-day course. In total, 75 students were exposed to a non-life-threatening simulation scenario before the intervention that included a personalized debriefing session as intervention, and followed by a life-threatening simulation scenario. Conversely, the other 71 students participated in a life-threatening simulation scenario before the intervention that included a personalized debriefing session as intervention, and followed by a non-life-threatening scenario after the intervention.

Educational theoretical background

The course design was based on Kolb's experiential learning theory, which posits that learning emerges from the transformation of experiences through four crucial stages: concrete experience, reflective observation, abstract conceptualization, and active experimentation [22]. In alignment with this theoretical approach, the simulation course was organized to start with a pre-intervention scenario, serving as the concrete experience. This was followed by a personalized debriefing session after the scenario, aimed at promoting reflective observation and abstract conceptualization. The course culminated in a post-intervention scenario designed for active experimentation.

The speaking-up training course

To evaluate the medical students' speaking-up behaviors and patterns during a medical error event, it was crucial to ensure they had adequate background knowledge to handle the clinical scenario presented to them. Two weeks prior to the simulation course, students were introduced that the training would involve diagnosing and treating a patient with paradoxical supraventricular tachycardia (PSVT) with online pre-course reading materials, including the American Heart Association (AHA) Guidelines for tachycardia.

On the day of the course, faculty members first conducted a 20-min review and discussion session to reinforce core knowledge on managing a PSVT patient. This was followed by a written test that students needed to pass before proceeding to the simulation scenario. Faculty offered guidance and instruction on the course material to students who did not pass the test on their first attempt. To ensure clarity and minimize any confusion regarding the simulation setup, students were briefed on the simulation environment, introduced to the standardized nurse responsible for executing their orders, and informed about their role in managing the simulated patient. They were not informed about the impending medical error event.

During the pre-intervention simulation scenario, each student interacted with a full-body computerized mannequin, a standardized nurse, and a standardized senior physician in scenarios with either stable or unstable vital signs, representing non-life-threatening and life-threatening situations, respectively. Once a student prescribed the correct medication, the standardized nurse proceeded to prepare it. Subsequently, the senior physician entered the simulation. Following a short interaction with the standardized nurse and the student, the senior physician would then mistakenly order the wrong medication for the patient. If the student remained silent about the medical error, the nurse would proceed to administer the incorrect medication. Conversely, if the student spoke up and voiced concern to the medical error, the senior physician would engage with the student to discuss their treatment recommendation. If the senior physician felt that students used offensive language, the senior physician would become defensive and insist on using the wrong medication.

Following the pre-intervention simulation scenario, a group of four students, who experienced the same scenario, participated in a faculty-led personalized debriefing session based on the gather, analyze, and summarize (GAS) model [23]. The session, lasting 60 min, was standardized through a scripted guide. It aimed at facilitating reflection, incorporating standardized speaking-up techniques, and helping students develop personalized speaking-up strategies, which they then practiced. The techniques introduced included the two-challenge rule, advocacy and assertion, and the CUS and DESC scripts from TeamSTEPPS [7]. After the personalized debriefing session, the students would attempt the post-intervention scenario.

After the post-intervention simulation scenario, a second 30-min debriefing session took place, where students and faculty discussed the application of personalized speaking-up strategies during the scenario and improvement based on their performance.

Assessment instruments and data collection

The outcome assessment of this study was to assess the patterns of medical students' speaking-up behavior when they encountered a medical error in a standardized scenario and the learning outcomes of the personalized simulated speaking-up training. All assessment tools used were discussed and revised by our expert panel, which included one simulation expert, two emergency physicians, two practicing nurse practitioners, and two practicing nurses. All the experts underwent formal simulation faculty development training and had at least 5 years of teaching experience in simulation. The students' speaking-up behaviors, pattens and confidence in the event of a medical error were assessed during both the pre-intervention and post-intervention scenarios. For the observational learning outcomes, two raters were trained as independent evaluators to assess students' speaking-up behavior in real-time scenarios, using standardized videos and discussions prior to the course.

Assessment of medical students' response to medical error

Trained raters evaluated response of medical students to medical errors by using a dichotomous scale marked 1 (speaking-up) and 0 (not speaking-up) checkbox for marking the students' response to the medical error event. Using a modified version of the Pian-Smith Evaluation Tool [24] – a 5-point rating scale where 1 indicates no reaction, 2 suggests negative facial or body language, 3 involves making an indirect or vague statement, 4 entails advocating or asking questions, and 5 represents persistent advocacy or inquiry, initiating discussion – our expert panel, including one simulation expert, two emergency attending physician, one nurse practitioner, and one registered nurse, defined a score of 4 or higher as clear speaking-up behavior in response to medical errors. Speaking up successfully in the event of a medical error was defined as doing so before the standardized nurse administered the incorrect medication ordered by a standardized senior physician. According to Krippendorff's alpha, the interrater reliability was 0.851.

Assessment of medical students' speaking-up patterns to medical error

To assess medical students' speaking-up patterns in the event of medical errors, our expert panel reviewed and discussed about the common clinical scenarios related to medical errors. Additionally, our expert panel shared and discussed their experiences of teaching speaking-up courses, with a focus on the sentences used by medical students. From these discussions, we then identified several key categories. The first category, the expression style of speaking-up pattern, assesses whether the medical students conveyed their concerns through direct expressions or indirect hints. The second category, the expression form of speaking-up pattern, examines whether the students communicated in affirmative sentences or posed question sentences. The third category, the attitude of speaking-up pattern, evaluates whether faculty members perceived the students' expressions as offensive or non-offensive. For instance, if a student says, 'I have asked the nurse to

Table 1. Participant demographics ($N = 146$).

Characteristics	Cohort
Age (year), mean (range)	23.56 (22–32)
Male/female *n* (%)	81 (55.5)/ 65 (44.5)
Clinical experience (month)	10.59 (6–16)
Received training course related to patient safety (Yes/ No), *n* (%)	119 (81.5) /24 (16.4)

prepare the medication for the patient,' this would be categorized as an indirect hint and an affirmative sentence. Conversely, if a student questions, 'Should we use adenosine as the first-line treatment?' it would be classified as a direct expression and a question sentence. If the raters perceive any words used by the participants as offensive to standardized senior physician during the simulated scenario, for example, if a participant says, 'This is your decision and has nothing to do with me,' it would be classified as offensive language. According to Krippendorff's alpha, the inter-rater reliability was 0.9 for style of speaking-up pattern, 0.734 for form of speaking up-pattern and 0.867 for attitude of speaking-up pattern.

Assessment of medical students' confidence of speaking up to medical errors

To assess the students' confidence of speaking up to medical errors before and after the intervention, our expert panel designed a 7-items questionnaire. The specific items in the questionnaire included: (1) 'I am confident in communicating with different healthcare professionals,' (2) 'I am confident in communicating with senior clinical practitioners,' (3) 'I am confident in my verbal communication skills,' (4) 'I am confident in my non-verbal communication skills,' (5) 'I am confident that I can adequately call for help when I encounter a difficult scenario,' (6) 'I am confident in providing safe healthcare for my patient,' and (7) 'In clinical practice, I am able to confidently apply "cultural awareness and sensitivity" in my interactions, which means I can adjust my approach to best suit the background and personality of the person I am dealing with at the moment.' 5-point Likert scale from 1 not at all to 5 almost always was applied to assess the confidence level. The Reliability Cronbach's alpha of this scale was 0.88.

Data analysis

We described the participants' demographic data related age and previous clinical experience using means with ranges. Gender and preciously participating in a patient safety course were reported using counts and percentages. We employed crosstab analysis for paired nominal data to evaluate changes in expression style, form, and attitude. We used Pearson's method for two-by-two contingency table, and applied NcMemar-Bowker method for two-by-three table due to three levels in expression style, expression form, and response attitude. Confidence survey responses were continuous data and was analyzed using Wilcoxon signed ranked test due to non-normal distribution. Statistical significance was defined as alpha <0.05 for all analyses in this study. These analyses were carried out using SPSS version 20.0 (IBM Corp., Armonk NY).

Table 2. Speaking-up rate to medical error in pretest scenario ($N = 146$).

			Crosstabs	
	Before debriefing	After debriefing	Pearson	*p*
Respond	64 (43.8%)	141 (96.6%)	6.633	0.015*
No respond	82 (56.2%)	5 (3.4%)		

Fisher's exam. $^*p < 0.05$, two-tailed.

Results

All 146 students, comprising 81 male and 65 female students, participated in and completed the course. The mean age of the students was 23.56 years (ranging from 22 to 32). Data on the speaking-up pattern of one student were missing and were excluded from the analysis. Additionally, all students completed the confidence survey. At the pre-intervention scenario, 64 students (43.8%) spoke up about the medical error, while at the post-intervention scenario, 141 students (96.6%) spoke up ($p = 0.015$).

The speaking-up patterns of students in the event of a medical error are presented in Table 3. A statistically significant change in expression style was observed between pre- and post-intervention scenario ($p < 0.001$). In the pre-intervention scenario, 50 students (34.4%) utilized direct expression, while 14 students (9.7%) used indirect hints. In the post-intervention scenario, 116 students (80%) employing direct expression and 24 students (16.6%) using indirect hints. Regarding expression form in the pre-intervention scenario, a statistically significant change in expression form was observed between pre- and post-intervention scenario ($p < 0.001$). In pre-intervention scenario, 31 students (21.4%) used affirmative sentences, whereas 33 students (22.7%) utilized questions. In the post-intervention scenario, 99 students (68.3%) employing affirmative sentences and 41 students (28.3%) using questions. In terms of speaking-up attitude in the pre-intervention scenario, a statistically significant change in expression attitude was observed between pre- and post-intervention scenario ($p < 0.001$). In the pre-intervention scenario, 47 students (32.4%) used unoffensive words, while 17 students (11.7%) used offensive words. In the post-intervention scenario, with 130 students (89.7%) using unoffensive words and 10 students (6.9%) using offensive words. Table 4 presents the details of speaking-up patterns across different scenarios. Before the intervention, the rate of speaking up among medical students was 50% in life-threatening scenarios and 38.7% in non-life-threatening scenarios. Following faculty-led personalized debriefing, the speaking-up rate increased to greater than 95% in both life-threatening and non-life-threatening scenarios (p value < 0.001). The students significantly adopted more direct (81.4% in life-threatening scenario first group and 78.7% in non-life-threatening scenario first group), affirmative (67.5% in life-threatening scenario first group and 70.7% in non-life-threatening scenario first group), and non-offensive (92.8% in life-threatening scenario first group and 86.7% in non-life-threatening scenario first group) speaking-up patterns.

Table 3. Medical students' expression methods in speaking up in response to medical errors in before/after debriefing.

Before debriefing		After debriefing ($N = 145$)			Crosstabs	
					NcMemar-Bowker	p
Expression style		Directly Expression ($n = 116$)	Indirect Hint ($n = 24$)	No respond ($n = 5$)	68.053	<0.001*
Directly Expression	50	39 (78.0%)	7 (14.0%)	4 (8.0%)		
Indirect Hint	14	11 (78.6%)	2 (14.3%)	1 (7.1%)		
No respond	81	66 (81.5%)	15 (18.5%)	0 (0.0%)		
Expression form		Affirmative ($n = 99$)	Question ($n = 41$)	No respond ($n = 5$)	72.920	<0.001*
Affirmative	31	22 (71.0%)	8 (25.8%)	1 (3.2%)		
Question	33	19 (57.6%)	10 (30.3%)	4 (12.1%)		
No respond	81	58 (71.6%)	23 (28.4%)	0 (0.0%)		
Attitude		Unoffensive ($n = 130$)	Offensive ($n = 10$)	No respond ($n = 5$)	76.145	<0.001*
Unoffensive	47	39 (83.0%)	3 (6.4%)	5 (10.6%)		
Offensive	17	16 (94.1%)	1 (5.9%)	0 (0.0%)		
No respond	81	75 (92.6%)	6 (7.4%)	0 (0.0%)		

Fisher's exam. *$p < 0.05$, two-tailed.

Table 4. Speaking-up response to medical error in before/after debriefing.

	Life-threatening ($N = 70$)		Crosstabs		Non-life-threatening ($N = 75$)		Crosstabs	
Respond way	Before debriefing ($n = 70$)	After debriefing ($n = 70$)	NcMemar-Bowker	p	Before debriefing ($n = 75$)	After debriefing ($n = 75$)	NcMemar-Bowker	p
Expression style			29.627	<0.001*			38.810	<0.001*
Direct Expression	29 (41.4%)	57 (81.4%)			21 (28.0%)	59 (78.7%)		
Indirect Hint	6 (8.6%)	11 (15.7%)			8 (10.7%)	13 (17.3%)		
No respond	35 (50.0%)	2 (2.9%)			46 (61.3%)	3 (4.0%)		
Expression form			30.493	<0.001*			43.855	<0.001*
Affirmative	19 (27.1%)	46 (65.7%)			12 (16.0%)	53 (70.7%)		
Question	16 (22.9%)	22 (31.4%)			17 (22.7%)	19 (25.3%)		
No respond	35 (50.0%)	2 (2.9%)			46 (61.3%)	3 (4.0%)		
Attitude			35.457	<0.001*			41.569	<0.001*
Unoffensive	28 (40.0%)	65 (92.8%)			19 (25.3%)	65 (86.7%)		
Offensive	7 (10.0%)	3 (4.3%)			10 (13.3%)	7 (9.3%)		
No respond	35 (50.0%)	2 (2.9%)			46 (61.3%)	3 (4.0%)		

Fisher's exam. *$p < 0.05$, two-tailed.

Students' confidence of speaking up in the event of medical errors significantly improved from a mean of 24.16 ± SD 4.16 in the pre-intervention survey to a mean of 27.16 ± SD 3.97 in the post-intervention survey ($p < 0.001$; see Appendix Table).

Discussion

The findings of this study suggest that simulated medical error scenario learning, followed by faculty-led personalized debriefing, can enhance students' speaking-up behavior. After the intervention, students utilized significantly more direct expression, affirmative sentences, and unoffensive words. Similar outcomes were observed in both life-threatening and non-life-threatening scenarios. This suggests that regardless of the order in which medical students experienced the scenarios, the faculty-led personalized debriefing positively influenced their speaking-up behavior. Additionally, their speaking-up confidence notably improved.

This study enhances understanding of speaking up in response to medical errors by delineating specific speaking-up patterns. This aspect has been insufficiently understood in previous research, despite evidence suggesting that training in assertive communication can improve students' speaking-up behavior [25–27]. This study contributes to the literature in two aspects. First, while previous literature may not have systematically structured speaking-up patterns, our study team identified potential key categories based on observations both in clinical experiences and medical education practices. Second, our study examines the style and form of expressions used by students before and after engaging in simulation scenario learning and faculty-led personalized debriefing. In acute care settings, particularly when patients are in a critical condition, healthcare providers have only a brief window of time to prevent medical errors from happening [28]. Using direct expression and affirmative sentences, healthcare providers can effectively and promptly address their concerns about medical errors. The findings of this study suggest that students are inclined to speak up in the event of medical errors using more direct expression and affirmative sentences, along with increased speaking-up confidence after the intervention. This indicates that simulation scenario learning and faculty-led personalized debriefing, supplemented with structured tools, can help students develop their speaking-up behavior, as evidenced by their altered patterns of addressing medical errors. Healthcare educators could use the intervention as a model to develop effective learning activities and prioritize assessing students' speaking-up patterns to achieve optimal learning outcomes.

Through various intensity scenarios and speaking-up patterns, our study may enhance understanding of medical students' behaviors in speaking up when confronting medical error events. In a previous study, we have revealed that medical students respond to medical errors differently depending on the clinical situation, such as whether the patient is in a life-threatening or non-life-threatening condition [10,11]. Instead of expressing their concerns about medical errors through direct and affirmative phrases, about 25% of students in this study express their concerns by asking questions. Some students feel psychologically safer and find it easier to express their concerns about medical errors by using indirect hints or posing questions [29]. For students, addressing concerns by asking questions may feel more like having a discussion or requesting

confirmation rather than challenging authority. Developing multiple speaking-up strategies to respond to different clinical scenarios would be valuable for students to utilize the most appropriate approach. Healthcare educators can focus more on discussing with students the advantages and disadvantages of various approaches of speaking-up in medical errors, helping them to develop effective speaking-up behaviors in a variety of medical contexts.

The findings of this study further reveal the diverse ways in which medical students respond to medical errors. Our data show that, after training, the majority of speaking-up patterns are direct, affirmative, and non-offensive, with about 20% of students using indirect hints or questions. However, these findings do not necessarily endorse a specific style of communication, as the appropriateness of speaking-up patterns is context-dependent. Our research team offers a range of speaking-up strategies, encouraging students to adapt these to the specific contexts of their encounters and their personal communication styles.

Healthcare providers' and students' intentions to speak up in the event of medical errors are influenced by culture and climate [30,31]. Medical students' concerns about speaking up may compromise their relationships with seniors and lead to repercussions [15]. It is important not only to encourage students to speak up in the event of medical errors but also to assist them in developing adequate and appropriate strategies to prevent negative outcomes. The results of this study indicate that faculty-led personalized debriefing sessions could encourage students to use unoffensive language when speaking up. This enables students to speak up with greater confidence without worrying about potentially offending their seniors. When conducting speaking-up courses, healthcare educators should not only focus on encouraging students to speak up but also on helping them develop appropriate language skills to facilitate speaking up. Healthcare educators should work closely with clinical practitioners and administrators to cultivate a safety culture in clinical settings, which facilitates the sharing of mental models and encourages speaking up. Along with training in speaking up, we can promote and ultimately achieve the goal of providing safe healthcare.

Providing psychologically safe environment is crucial for students to learn in simulation [32]. Despite faculties' efforts to create a safe environment in simulation settings, some students may still feel stressed about addressing their concerns or struggle to apply the strategies they have learned for addressing medical errors. Identifying these students and providing further assistance are critical tasks for healthcare educators. Future studies should explore who may require additional assistance even before the intervention, allowing educators to offer more accurate and effective personalized learning activities.

Limitations

The generalizability of the study is restricted due to its single-site nature, conducted solely at a medical university in Northern Taiwan, potentially limiting applicability to broader contexts with diverse cultural and educational settings. The sample size of 146 medical students, primarily consisting of senior medical students, might not adequately represent the wider population of healthcare professionals or other healthcare professions students at varying training stages. While simulation scenarios offer controlled environments for observing behaviors, they may not entirely mirror real clinical settings where medical errors typically occur in unexpected situations. This discrepancy could potentially influence the students' responses and behavior differently in actual patient care scenarios. Additionally, the subjectivity of speaking-up patterns, including style, form, and attitude, may depend on inter-rater reliability.

Conclusions

This study demonstrates that the combination of simulated medical error scenarios and faculty-led personalized debriefing sessions could enhance and modify medical students' speaking-up behavior in the event of medical errors. Following this intervention, students demonstrated a significant increase in using direct expression, affirmative sentences, and non-offensive language when addressing medical errors, alongside a notable improvement in their confidence. This study does not emphasize a specific style or form of expression but highlights the importance of personalized debriefing, which can inspire medical students to develop their own strategies for responding to medical errors. In addition, this study acknowledges the necessity of combining simulation-based training with personalized debriefing in shaping students' behaviors toward medical errors. Healthcare educators may consider developing and refining strategies to foster assertive communication skills among students, thereby advancing patient safety and care quality.

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Author contributions

Hui-Wen Chen, Yi-No Kang, and Jen-Chieh Wu: conceptualization. Hui-Wen Chen, Yi-No Kang: formal analysis. Yi-Chun Chen and Hung-Wei Tsai: investigation. Yi-No Kang: methodology. S. Barry Issenberg and Jen-Chieh Wu: supervision. Hui-Wen Chen and Jen-Chieh Wu: writing – original draft. Hung-Wei Tsai, S. Barry Issenberg, Yi-Chun Chen, Enoch Yi-No Kang: writing – review and editing.

Disclosure statement

Hung-Wei Tsai, S. Barry Issenberg, Yi-Chun Chen, Enoch Yi-No Kang, Hui-Wen Chen, and Jen-Chieh Wu states that they have no financial or non-financial conflicts of interest to disclose related to this manuscript.

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